



Final Exam, Theory of Computation 2025

1 (15 pts) **Quick-fire round.** Consider the following statements.

1. If $A \leq_p B$, then $A \leq_m B$.
2. If $L \leq_p \text{SAT}$, then $L \leq_p \text{CLIQUE}$.
3. If $L \subseteq \text{SAT}$ then $L \in \mathbf{NP}$.
4. If $\text{SAT} \notin \mathbf{coNP}$, then $\mathbf{NP} \neq \mathbf{coNP}$.
5. If $\text{GRAPH-ISOMORPHISM} \in \mathbf{P}$, then $\mathbf{P} = \mathbf{NP}$.
6. If L is recognisable, then $L \leq_m \text{HALT}$.
7. If L is decidable, then L^* is decidable.
8. If $A \cup B$ is recognisable, then either A or B is recognisable.
9. Every boolean circuit can be equivalently written as a CNF formula.
10. Language $\{0^n 1^m : n \neq m\}$ is regular.
11. Language $\{x \in \{0,1\}^* : x \text{ has the same number of occurrences of } 01 \text{ and } 10\}$ is regular.
12. Language $\{\langle \varphi \rangle : \varphi \text{ is a CNF that has } \geq 2025 \text{ satisfying assignments}\}$ is $\mathbf{NP}$ -complete.
13. Language $\{\langle G, k \rangle : G \text{ is a graph with no vertex cover of size } k\}$ is $\mathbf{NP}$ -complete.
14. Language $\{\langle M \rangle : M \text{ is a TM and } L(M) \text{ is regular}\}$ is decidable.
15. Language $\{\langle n, m \rangle : \text{BUSYBEAVER}(n) \geq m\}$ is recognisable.

For each statement, write down one of

- **T** if the statement is known to be true.
- **F** if the statement is false *or not known to be true*. E.g., both $\mathbf{P} = \mathbf{NP}$ and $\mathbf{P} \neq \mathbf{NP}$ should be marked **F**.
- or leave your answer empty.

A correct **T/F** answer is worth +1 point, an incorrect answer is worth −1 point, and an empty answer is worth 0 points.

2 (15 pts) Regular languages I.

2a Write down the statement of the Pumping Lemma. (*Only the statement, not its proof!*)

2b Consider strings over the alphabet $\Sigma = \{[,]\}$ and the language

$$L = \{x \in \Sigma^* : x \text{ is well-nested}\}.$$

For example, $[] [] \in L$, $[[] [[]]] \in L$, but $[]] \notin L$. Show that L is not regular.

3 (15 pts) NP-completeness I. A *shy set* in an undirected graph is a subset S of nodes such that each node in S has at most one neighbour in S . Show that SHY is **NP**-complete:

$$\text{SHY} = \{ \langle G, k \rangle : G \text{ is a graph that has a shy set of size } k \}.$$

4 (20 pts) NP-completeness II. Prove that SAT remains **NP**-complete under each of the following conditions on the input:

4a No clause contains both a positive literal and a negative literal.
(*Hint: Use a 2-CNF to express that two variables are assigned opposite bits.*)

4b No variable occurs (positively or negatively) in more than three clauses.

5 (20 pts) Undecidability. Consider the language

$$\text{EXP} = \{ \langle M \rangle : M \text{ is a TM that halts in time } \leq 2^{|x|} \text{ on any input } x \in \{0, 1\}^* \}.$$

Here $|x|$ denotes the length of the string x . Classify EXP as one of (i) decidable, (ii) undecidable but recognisable, (iii) unrecognisable. Justify your answer with a proof.

6 (15 pts) Regular languages II. Let $|x|_b$ denote the number of occurrences of bit $b \in \{0, 1\}$ in a binary string $x \in \{0, 1\}^*$. For every $n \geq 2$, define $L_{n,b} = \{x \in \{0, 1\}^* : |x|_b \bmod n = 0\}$.

6a Describe a DFA with n states that accepts $L_{n,b}$.

6b Show that any NFA accepting $L_{n,0} \cap L_{n,1}$ has at least n^2 states.
(*Hint: If an NFA uses fewer than n^2 states, use a pigeonhole argument to find two (cleverly chosen) strings that lead to the same state.*)